

HOSPITAL EMERGENCY ROOM

ANALYSIS REPORT

Patient Flow & Operational Performance Analysis

1 April 2023 – 30 October 2024

1. Business Problem

Emergency departments need to balance unpredictable patient demand with limited staff, treatment capacity, and specialist resources. A high patient flow can create longer waits, delays in care, pressure on referral departments, and weaker patient experience.

Key analytical questions

- How does patient demand change over time across 2023 and 2024?
- What proportion of emergency-room visits result in admission?
- Which hours, weekdays, months, and dates show higher patient volumes?
- How long are patients waiting, and what share is seen within 30 minutes?
- Which departments receive the largest number of referrals?
- How are patients distributed by gender, age group, and race?
- What does the satisfaction data indicate, and how should zero scores be interpreted?

2. Executive Summary

The emergency room dataset contains 9,216 patient visit records covering 01 April 2023 to 30 October 2024. The analysis focuses on patient demand, admission outcomes, waiting time, department referrals, patient satisfaction, and demographic patterns.

Total Visits	Admitted	Average Wait	Within 30 Min	Delayed	Avg Satisfaction*
9,216	4,612 (50.0%)	35.3 min	40.7%	59.3%	1.36/10

The main operational signal is that demand is fairly steady but service performance shows room for improvement: the average waiting time is 35.3 minutes, only 40.7% of visits are completed within 30 minutes, and 59.3% of records are classified as delayed. General Practice is the largest recorded referral destination, while a large share of visits has no department referral recorded.

Across the two calendar years represented, 2024 contains more visits than 2023 in absolute terms. However, the periods are not equal: 2023 begins in April, while 2024 extends through October. For this reason, monthly averages are more meaningful than a simple full-year growth comparison.

*The satisfaction field contains a large number of zero values. The report therefore presents both the overall field average and a non-zero-score view to avoid treating potentially unrecorded survey responses as genuine satisfaction ratings.

4. Core Performance Findings

4.1 Patient volume and admission

There are 9,216 visit records and 9,185 distinct names in the Name field. Admissions account for 4,612 visits, giving an overall admission rate of 50.04%. The near 50/50 split between admitted and non-admitted visits indicates that the emergency room is serving a substantial mix of cases requiring both hospital admission and emergency assessment without admission.

Year	Visits	Admitted	Admission Rate	Avg Wait	Within 30 Min	Avg Visits / Month
2023	4,338	2,157	49.7%	35.0	42.0%	15.8
2024	4,878	2,455	50.3%	35.5	39.5%	16.0

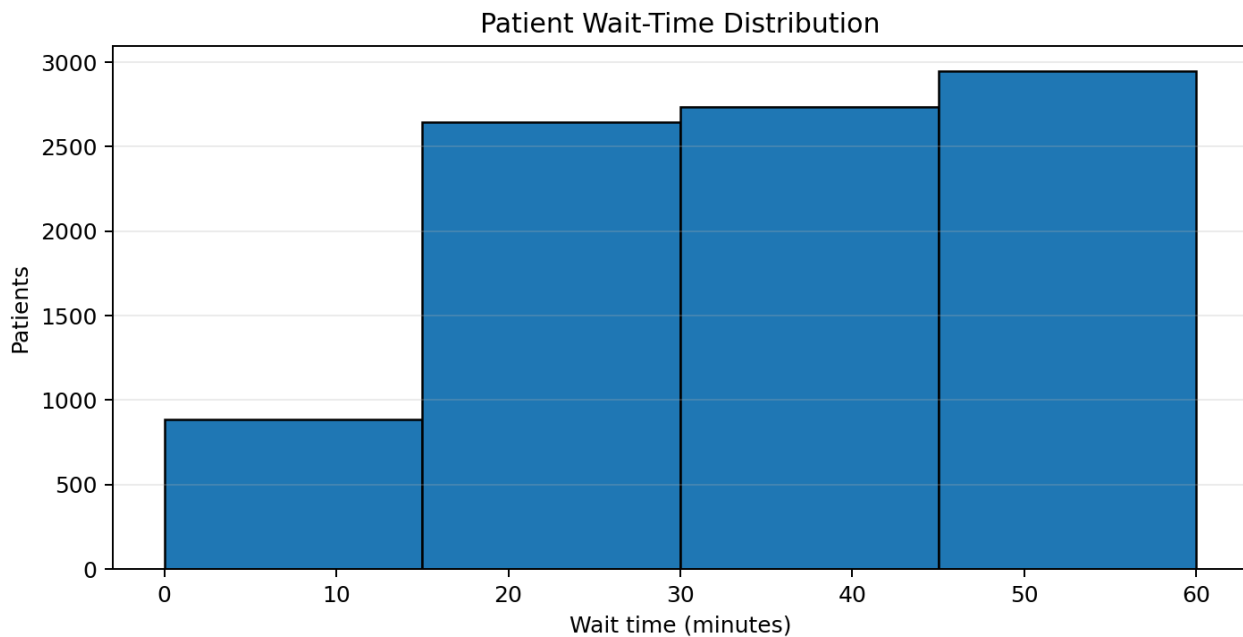
2024 has 4,878 visits compared with 4,338 in 2023. Because the observation windows differ, this should not be interpreted as a like-for-like annual growth rate. The average monthly volume is 15.8 in 2023 and 16.0 in 2024, a change of 1.7%.

4.2 Waiting time and service delay

Average waiting time is 35.3 minutes, with a maximum of 60 minutes. Only 40.7% (3,749 visits) were recorded at 30 minutes or less, while 59.3% of records are classified as delayed.

Wait Time	Patients	Share
0–15	1,082	11.7%
16–30	2,667	28.9%
31–45	2,717	29.5%
46–60	2,750	29.8%

The distribution shows that the largest group falls in the 46–60 minute range. This indicates that the waiting-time issue is not limited to a small number of extreme cases; a meaningful share of visits experience waits above 30 minutes.



4.3 Patient demand by time

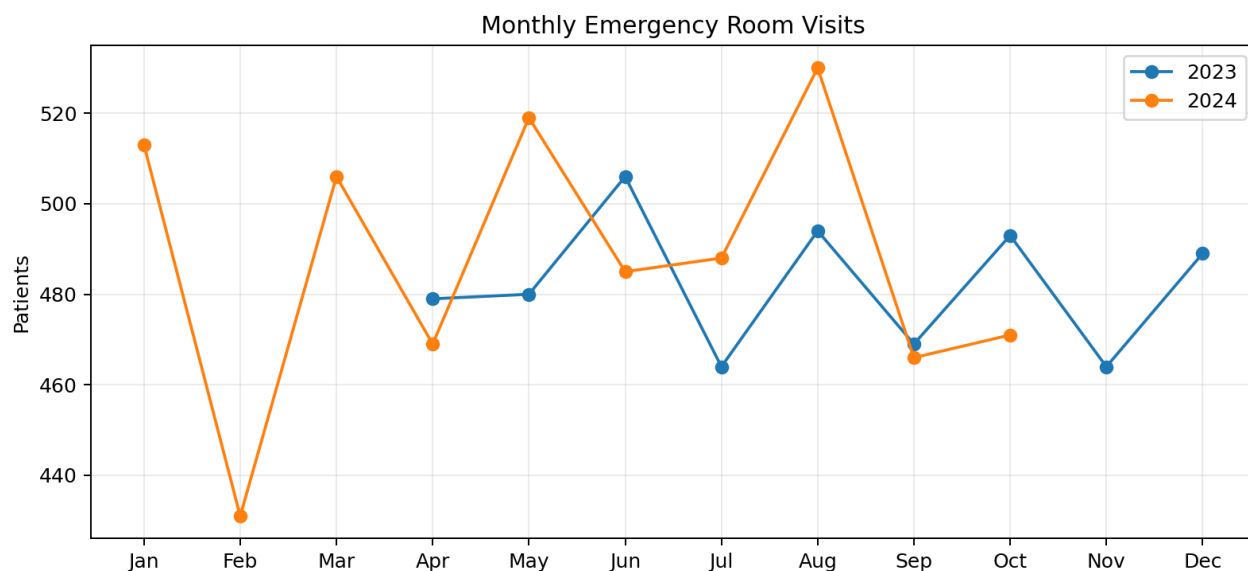
Saturday is the busiest weekday with 1,377 visits (14.9% of all visits), while Wednesday records the lowest volume at 1,260 (13.7%). The highest-volume hour in the dataset is 23:00–23:59 with 436 visits.

Rank	Hour	Visits	Share
1	23:00–23:59	436	4.7%
2	07:00–07:59	415	4.5%
3	13:00–13:59	410	4.4%
4	00:00–00:59	406	4.4%
5	11:00–11:59	403	4.4%

The hourly pattern should be used with staffing and shift planning rather than as a standalone indicator. Demand is distributed across the full day, so operational coverage should account for sustained activity rather than only one peak hour.

4.4 Monthly and peak-day demand

Across the available months, August has the highest combined volume, followed by May and June. At the individual-date level, the busiest recorded day has 30 visits.



Rank	Date	Visits
1	12 Apr 2023	30
2	21 May 2024	28
3	01 Aug 2024	28
4	15 Dec 2023	27
5	22 Jun 2023	26
6	07 Mar 2024	26
7	15 Sep 2023	26
8	01 Jun 2023	26
9	17 Aug 2024	25
10	12 Jul 2024	25

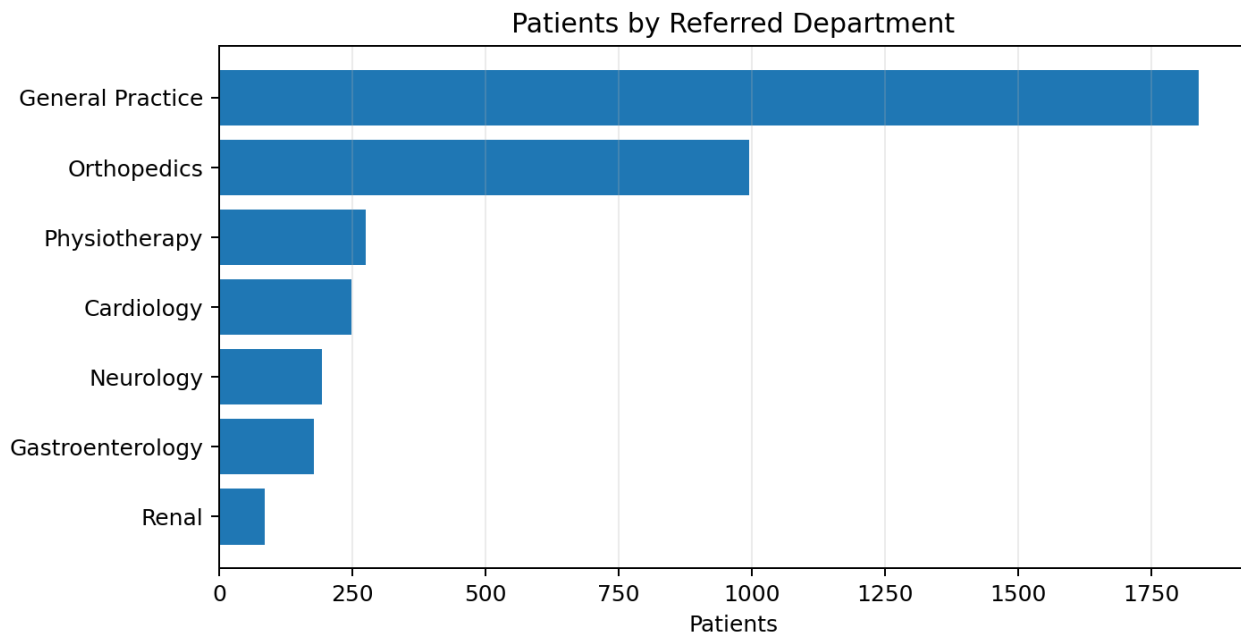
5. Department Referral Analysis

Department referral is recorded for 3,816 visits (41.4% of all records). The remaining 5,400 records (58.6%) have no referral department recorded.

Department	Patients	Share of All Visits	Share of Recorded Referrals
General Practice	1,840	20.0%	48.2%
Orthopedics	995	10.8%	26.1%
Physiotherapy	276	3.0%	7.2%
Cardiology	248	2.7%	6.5%
Neurology	193	2.1%	5.1%

Gastroenterology	178	1.9%	4.7%
Renal	86	0.9%	2.3%

General Practice is the largest recorded referral destination with 1,840 visits, representing 20.0% of all visits and 48.2% of recorded referrals. Orthopedics is second with 995 visits.



The high volume of blank department referrals is an important data-quality and process point. Improving referral capture would make department-level workload analysis more complete and would support better capacity planning.

6. Demographic Profile

The patient mix is relatively balanced by gender: males account for 51.3% and females 48.7%.

Age Group	Patients	Share
20-29	1,207	13.1%
30-39	1,191	12.9%
0-9	1,176	12.8%
10-19	1,160	12.6%
60-69	1,150	12.5%
50-59	1,147	12.4%
40-49	1,137	12.3%
70-79	1,048	11.4%

The age distribution is broad, with each age group representing a similar share of total visits. This suggests that the emergency-room workload is not concentrated in a single age segment.

Race	Patients	Share
White	2,571	27.9%
African American	1,951	21.2%
Two or More Races	1,557	16.9%
Asian	1,060	11.5%
Declined to Identify	1,030	11.2%
Pacific Islander	549	6.0%
Native American/Alaska Native	498	5.4%

Demographic results are descriptive only. They should be used to understand service demand and population coverage, not to infer clinical need or outcomes for individual groups.

7. Patient Satisfaction

The raw satisfaction field has an overall average of 1.36/10. However, 6,921 records (75.1%) contain a score of 0. Among the 2,295 records with a non-zero score, the average is 5.47/10.

This difference is important for interpretation. If a score of 0 represents an unrecorded or unavailable survey rather than true dissatisfaction, the overall average should not be used as the sole measure of patient experience. A useful dashboard improvement would be to report survey response coverage separately and calculate satisfaction only from valid responses.

Department	Average Satisfaction Non-Zero
Gastroenterology	5.91/10
Neurology	5.60/10
Physiotherapy	5.59/10
General Practice	5.57/10
Orthopedics	5.48/10
Cardiology	5.45/10
Renal	5.25/10

8. Analytical Insights

- Demand is steady: average monthly visits are 15.8 in 2023 and 16.0 in 2024, despite the different observation windows.
- Admission is almost evenly split, with an overall admission rate of 50.0%.

- Waiting time is a major operational indicator: the average is 35.3 minutes and only 40.7% of visits are within 30 minutes.
- The 46–60 minute waiting band contains 2,750 visits (29.8%), making longer waits a broad operational pattern rather than an isolated issue.
- General Practice and Orthopedics account for 30.8% of all visits with a recorded referral destination.
- Saturday has the highest weekday volume, indicating that staffing plans should consider weekend demand rather than focusing only on weekdays.
- Department referral data is incomplete for 58.6% of visits, limiting the completeness of department workload analysis.
- Satisfaction data needs a data-quality check because 75.1% of records have a zero score.

9. Data-Driven Operational Recommendations

Align staffing with demand: Use hourly, weekday, monthly and peak-day patterns to review staff coverage during higher-volume periods, especially weekends and identified peak hours.

Reduce waits above 30 minutes: Break the patient journey into registration, triage, clinical assessment and referral steps to identify where time accumulates. Track the percentage within 30 minutes as a recurring KPI.

Monitor sustained delay: Because a large share of visits falls above 30 minutes, monitor delay performance by hour, day and department rather than focusing only on maximum wait time.

Improve referral data capture: Investigate why department referral is blank for 58.6% of visits. A more complete referral field will improve department workload and capacity planning.

Improve satisfaction measurement: Confirm whether zero satisfaction values mean no survey response or an actual score. Report survey response rate and valid-score satisfaction separately.

Use comparable year-over-year views: For future reporting, compare the same calendar months across years or use rolling 12-month measures to avoid misleading comparisons caused by partial-year coverage.

10. Recommended Management Dashboard View

Dashboard Area	Primary KPI	Suggested management question
Demand	Visits / day and month	When is demand highest?
Flow	Average wait; % within 30 min	Where is patient flow slowing?
Admission	Admission rate	How much demand converts to inpatient care?
Referral	Referral volume by department	Which departments receive the most workload?

Experience	Valid satisfaction score; response rate	How is patient experience changing?
Capacity	Peak day / peak hour	Where should staffing be reviewed?
Data quality	Missing referral and satisfaction coverage	Which fields need better capture?

11. Conclusion

The emergency-room analysis provides a clear view of patient demand and operational performance across the available period. The service handled 9,216 visits with an admission rate close to 50%, while average waiting time remained at 35.3 minutes and 40.7% of visits were within 30 minutes. Demand patterns vary by month, weekday and hour, creating opportunities to align staffing and workflow with actual patient flow.

The analysis also highlights two measurement priorities: improving the completeness of department referral information and clarifying the meaning of zero satisfaction scores. Addressing these data-quality issues would make future performance reporting more reliable and would strengthen the hospital's ability to connect patient demand with operational capacity and patient experience.